Analyzing MLB Statistics With SQL

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Overview

Goal: Analyze MLB player statistics using SQL to identify trends in player and team performance

Why: I have always been interested in the analytical side of the sport. Working with data is something I am passionate about, and I find it fascinating how statistics can reveal trends in player and team performance while helping answer important questions about the game.

Questions Explored:

- Highest Batting Average Players
- Home Run Leaders
- Team Home Run Totals
- Stolen Base Leaders
- Highest WAR Players
- Highest OBP Players

Data Collection and SQL

- Extracted CSV files for every MLB team from Baseball Reference
- Cleaned each CSV file to keep only important statistics
- Fixed formatting issues such as player names with accent marks
- Combined all team CSV files into one complete MLB dataset
- Imported the final dataset into SQL for analysis
- **SQL Environment:** DataLab from Data Camp
- **DataBase Type:** Designed for OLAP, focuses on analyzing the trends and player performance.
- The data was cleaned and organized to reduce redundant information and make SQL queries easier to run efficiently.
- SQL was a good fit for this project because baseball produces large amounts of structured statistical data that can be efficiently filtered, grouped, and analyzed.

Baseball Statistics

- Batting Average (**BA**): How often a player gets a hit.
- **Formula:** Hits/At Bats
- **Ex**. 5/25 = .200

- Home Runs (**HR**): How many times a player hits the ball over the fence in the outfield for an automatic Score.

- On Base Percentage (**OBP**): How often a player reaches base successfully without making an out
- **Formula:** H + BB + HBP/AB + BB + HBP + SF
- **Ex**. 150 + 60 + 5/500 + 60 + 5 + 4 = 0.378

- Stolen Base (**SB**): Shows how often a player successfully steals the next base.

- Wins Above Replacement (**WAR**): Estimates a player's total overall value.

MLB_player

D1 | pos

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Rk	Player	Age	pos	WAR	G	PA	AB	R	H	2B	3B	HR	RBI	SB	CS	BB	SO
2	1	Cal Raleigh	28	C	7.4	159	705	596	110	147	24	0	60	125	14	4	97	188
3	2	Josh Naylor	28	1B	2.2	54	210	194	32	58	10	0	9	33	19	0	11	34
4	3	Cole Young	21	2B	0.5	77	257	223	24	47	7	1	4	24	1	0	28	47
5	4	J.P. Crawford	30	SS	3.8	157	654	570	69	151	24	0	12	58	8	1	74	122
6	5	Ben Williamson	24	3B	1.3	85	295	277	36	70	13	0	1	21	5	1	15	64
7	6	Randy Arozarena	30	LF	4	160	709	613	95	146	32	1	27	76	31	6	64	191
8	7	Julio Rodríguez	24	CF	6.8	160	710	652	106	174	31	4	32	95	30	6	44	152
9	8	Dominic Canzone	27	RF	1.6	82	269	243	30	73	11	0	11	32	3	1	20	59
10	9	Jorge Polanco	31	DH	2.6	138	524	471	64	125	30	0	26	78	6	3	42	82
11	10	Mitch Garver	34	C	-0.1	87	290	254	29	53	5	1	9	30	3	0	30	80
12	11	Eugenio Suárez	33	3B	0.3	53	220	201	27	38	9	0	13	31	3	1	17	79
13	12	Luke Raley	30	UT	-0.3	73	219	183	23	37	8	0	4	19	2	1	19	64
14	13	Dylan Moore	32	UT	0.2	88	213	192	29	37	5	0	9	19	12	4	19	76
15	14	Rowdy Tellez	30	1B	-0.8	62	185	173	20	36	6	0	11	27	1	0	8	49
16	15	Donovan Solano	37	1B	0	69	176	163	10	41	4	1	3	21	0	0	8	38
17	16	Miles Mastrobuoni	29	UT	0.2	76	175	152	20	38	4	0	1	12	6	3	17	29
18	17	Víctor Robles	28	RF	-0.2	32	114	106	12	26	4	1	1	9	6	2	3	26
19	18	Leo Rivas	27	2B	0.8	48	111	90	19	22	2	0	2	9	6	0	20	24
20	19	Leody Taveras	26	RF	-0.6	28	98	92	6	16	3	0	2	9	3	2	3	27
21	20	Ryan Bliss	25	2B	-0.2	11	39	35	1	7	1	0	1	3	2	2	4	11

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Workbook Statistics

MLB_player

D1 | pos

	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI
1	BA	OBP	SLG	OPS	OPS+	rOBA	Rbat+	TB	GIDP	HBP	SH	SF	IBB	K_rate	BB_rate	HR_rate	Team
2	0.247	0.359	0.589	0.948	166	0.392	165	351	6	9	0	3	17	0.2667	0.1376	0.0851	Mariners
3	0.299	0.341	0.49	0.831	135	0.38	141	95	3	2	0	1	0	0.1619	0.0524	0.0429	Mariners
4	0.211	0.302	0.305	0.607	75	0.273	77	68	4	2	2	2	1	0.1829	0.1089	0.0156	Mariners
5	0.265	0.352	0.37	0.722	108	0.322	113	211	11	3	6	1	1	0.1865	0.1131	0.0183	Mariners
6	0.253	0.294	0.31	0.604	74	0.272	75	86	5	1	2	0	0	0.2169	0.0508	0.0034	Mariners
7	0.238	0.334	0.426	0.76	116	0.341	124	261	9	27	0	5	2	0.2694	0.0903	0.0381	Mariners
8	0.267	0.324	0.474	0.798	125	0.353	133	309	15	12	0	2	5	0.2141	0.062	0.0451	Mariners
9	0.3	0.358	0.481	0.84	138	0.361	142	117	6	3	0	2	0	0.2193	0.0743	0.0409	Mariners
10	0.265	0.326	0.495	0.821	131	0.348	134	233	7	3	3	5	1	0.1565	0.0802	0.0496	Mariners
11	0.209	0.297	0.343	0.639	83	0.286	86	87	6	3	0	3	1	0.2759	0.1034	0.031	Mariners
12	0.189	0.255	0.428	0.682	91	0.289	89	86	4	1	0	1	1	0.3591	0.0773	0.0591	Mariners
13	0.202	0.319	0.311	0.631	83	0.294	94	57	2	13	3	1	0	0.2922	0.0868	0.0183	Mariners
14	0.193	0.263	0.359	0.622	76	0.283	83	69	1	0	0	2	0	0.3568	0.0892	0.0423	Mariners
15	0.208	0.249	0.434	0.682	90	0.289	89	75	6	2	0	2	0	0.2649	0.0432	0.0595	Mariners
16	0.252	0.295	0.344	0.639	83	0.284	87	56	9	3	0	2	0	0.2159	0.0455	0.017	Mariners
17	0.25	0.324	0.296	0.62	80	0.279	82	45	2	0	5	1	0	0.1657	0.0971	0.0057	Mariners
18	0.245	0.281	0.33	0.611	75	0.282	82	35	3	3	0	2	0	0.2281	0.0263	0.0088	Mariners
19	0.244	0.387	0.333	0.721	110	0.343	120	30	1	1	0	0	0	0.2162	0.1802	0.018	Mariners
20	0.174	0.198	0.272	0.47	33	0.21	33	25	0	0	2	1	0	0.2755	0.0306	0.0204	Mariners
21	0.2	0.282	0.314	0.596	71	0.253	73	11	2	0	0	0	0	0.2821	0.1026	0.0256	Mariners

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Workbook Statistics

Query #1: Highest BA (Batting Average)

- Which players have the highest BA?

- This query finds the players with the highest BA (Batting Average).

- I filtered for players with more than 350 at-bats to avoid small sample sizes affecting the results. This allows the analysis to focus on everyday players rather than bench players with limited playing time.

```sql
SELECT Player, Team, BA
FROM player.csv
WHERE AB > 350
ORDER BY BA DESC
LIMIT 15;
```

DataFrames and CSVs | DataFrame available as df1

Table Chart Filter Columns Search

index	Player	Team	BA	AB
0	Aaron Judge	Yankees	0.331	541
1	Jonathan Aranda	Rays	0.316	370
2	Bo Bichette	Blue_Jays	0.311	582
3	Jacob Wilson	Athletics	0.311	486
4	George Springer	Blue_Jays	0.309	498
5	Jeremy Peña	Astros	0.304	493
6	Trea Turner	Phillies	0.304	589
7	Yandy Díaz	Rays	0.3	583
8	Nico Hoerner	Cubs	0.297	599
9	Will Smith	Dodgers	0.296	362
10	Jake Mangum	Rays	0.296	405
11	Chandler Simpson	Rays	0.295	414
12	Freddie Freeman	Dodgers	0.295	556
13	Bobby Witt Jr.	Royals	0.295	623
14	Vladimir Guerrero Jr.	Blue_Jays	0.292	589

Rows: 15

Query #2: Home Run Leaders

- This query explores the top Home Run hitters in the league

- Home runs are seen as one of baseball's most valuable offensive statistics because they guarantee at least one run scored.

- Players with more home runs contribute heavily to scoring and offensive production.

```
SELECT Player, Team, HR
FROM player.csv
ORDER BY HR DESC
LIMIT 10;
```

DataFrames and CSVs   DataFrame available as df2

Table   Chart   Filter   Columns   Search

index	Player	Team	HR
0	Cal Raleigh	Mariners	60
1	Kyle Schwarber	Phillies	56
2	Shohei Ohtani	Dodgers	55
3	Aaron Judge	Yankees	53
4	Junior Caminero	Rays	45
5	Juan Soto	Mets	43
6	Pete Alonso	Mets	38
7	Jo Adell	Angels	37
8	Taylor Ward	Angels	36
9	Nick Kurtz	Athletics	36

Query #3: Team HR Totals

- Exploring which teams hit the most Home Runs

- This query groups players by team and calculates the total number of home runs for each organization. It helps compare which teams rely most heavily on power.

- Teams with higher home run totals often rely on power hitting to score runs and create offense.

```
SELECT Team,
       SUM(HR) AS Total_Home_Runs
FROM player.csv
GROUP BY Team
ORDER BY Total_Home_Runs DESC;
```

	Team	Total_Home_Runs
0	Yankees	274
1	Dodgers	244
2	Mariners	238
3	Angels	226
4	Mets	224
5	Cubs	223
6	Athletics	219
7	DiamondBacks	214
8	Phillies	212
9	Tigers	198
10	Twins	191
11	Blue_Jays	191
12	Orioles	191
13	Braves	190
14	Red_Sox	186
15	Rays	182
16	Astros	182
17	Rangers	175
18	Giants	173
19	Guardians	168
20	Reds	167
21	Brewers	166
22	White_Sox	165
23	Rockies	160
24	Royals	159
25	Marlins	154
26	Padres	152
27	Cardinals	148
28	Pirates	117

Query #4: Stolen Base Leaders

```
SELECT Player, Team, SB
FROM player.csv
ORDER BY SB DESC
LIMIT 10;
```

- Who are the most aggressive and successful base runners?

- This query helps us identify the players with the most stolen bases, stolen bases rely on timing, speed and awareness.

- Players with a higher count of stolen bases tend to create a different type of offensive pressure than compared to home runs.

	Player	Team	SB
0	Chandler Simpson	Rays	44
1	José Ramírez	Guardians	44
2	Juan Soto	Mets	38
3	Bobby Witt Jr.	Royals	38
4	Oneil Cruz	Pirates	38
5	Elly De La Cruz	Reds	37
6	Trea Turner	Phillies	36
7	Pete Crow-Armstrong	Cubs	35
8	José Caballero	Rays	34
9	Victor Scott II	Cardinals	34

Query #5: Highest WAR Players

```
SELECT Player, Team, WAR
FROM player.csv
ORDER BY WAR DESC
LIMIT 10;
```

- Which players provide the most value to their team?

- This query explores which players contributed the most overall value to their teams by analyzing their WAR

- WAR combines multiple factors such as offense, defense, and base running into one statistic

	Player	Team	WAR
0	Aaron Judge	Yankees	9.7
1	Cal Raleigh	Mariners	7.4
2	Bobby Witt Jr.	Royals	7.1
3	Geraldo Perdomo	DiamondBacks	7
4	Julio Rodríguez	Mariners	6.8
5	Shohei Ohtani	Dodgers	6.6
6	Juan Soto	Mets	6.2
7	Nico Hoerner	Cubs	6.2
8	Corey Seager	Rangers	6.2
9	Pete Crow-Armstrong	Cubs	6

Query #6: Highest OBP Players

- Which players have reached base most consistently?

- This query helps us identify players who are most effective at consistently reaching base.

- OBP (On-Base Percentage) measures how often a player reaches base through hits, walks, or being hit by a pitch.

```
SELECT Player, Team, OBP
FROM player.csv
WHERE PA > 350
ORDER BY OBP DESC
LIMIT 10;
```

	Player	Team	OBP
0	Aaron Judge	Yankees	0.457
1	Ronald Acuña Jr	Braves	0.417
2	Will Smith	Dodgers	0.404
3	George Springer	Blue_Jays	0.399
4	Juan Soto	Mets	0.396
5	Jonathan Aranda	Rays	0.393
6	Shohei Ohtani	Dodgers	0.392
7	Geraldo Perdomo	DiamondBacks	0.389
8	Nick Kurtz	Athletics	0.383
9	Vladimir Guerrero Jr.	Blue_Jays	0.381

What I learned

- Learned how to clean, organize, and analyze large datasets using SQL
- Improved understanding of how sports analytics uses statistics to evaluate player and team performance
- Learned how filtering and sample size can impact the accuracy of statistical analysis
- Discovered how SQL can transform raw data into meaningful insights and trends

Conclusion

- SQL is an effective tool for analyzing large sports datasets
- Baseball statistics can reveal meaningful trends in player and team performance
- Advanced metrics like WAR and OBP provide deeper insight than basic statistics alone
- Data analytics helps organizations make more informed decisions
- This project demonstrated how SQL and data analytics can transform raw baseball statistics into meaningful insights about player and team performance

Works Cited

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